26900192 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

FIFTH SEMESTER

Discipline Specific Elective (DSE) - MG5DSEECC300 - MICROWAVE ELECTRONICS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Which device is commonly used as a microwave amplifier? [K] / [CO1]

   a). Klystron b). Transformer

   c). Relay d). Rectifier bridge

2. In a TE mode, the electric field component along the direction of propagation is: [K] / [CO2]

   a). Zero b). Maximum

   c). Infinite d). Constant

3. The process in which variations in electron velocity are converted into variations in electron density is called: [U] / [CO3]

   a). Velocity modulation b). Density modulation or current modulation

   c). Frequency modulation d). Phase cancellation

4. An important parameter used to evaluate an isolator is: [U] / [CO4]

a). Reverse isolation b). Battery capacity

c). DC resistance only d). Transistor beta

5. Which electromagnetic waves are commonly used for broadcasting radio and television signals? [K] / [CO1]

a). Radio waves b). X-rays

c). Gamma rays d). Ultraviolet rays

6. Which parameters determine the propagation characteristics of a transmission line? [U] / [CO2]

a). Voltage only b). Current only

c). Load resistance only d). R, L, G and C

7. The interaction between the electron cloud and the RF fields in the cavities results in: [U] / [CO3]

a). Microwave oscillations b). DC amplification only

c). Heat generation only d). Optical radiation

8. A properly designed waveguide bend should ideally have: [U] / [CO4]

a). Low reflection and low insertion loss b). Maximum reflection

c). Maximum radiation d). Zero power transmission

9. Which type of loss is particularly important in coaxial cables operating at microwave frequencies? [U] / [CO2]

a). Conductor loss due to skin effect b). DC loss only

c). Magnetic saturation loss d). Transformer loss

10. Conventional vacuum tubes operate by controlling the flow of: [K] / [CO3]

a). Holes b). Electrons

c). Neutrons d). Photons

11. In a TE mode, which field component along the direction of propagation is zero? [K] / [CO1]

a). Electric field b). Magnetic field

c). Both electric and magnetic fields d). Neither field

12. If the operating frequency is increased significantly above cutoff, the group velocity: [U] / [CO2]

a). Approaches the velocity of light b). Approaches zero

c). Becomes infinite d). Remains unchanged

13. A reflex klystron uses how many resonant cavities? [K] / [CO3]

a). One b). Two

c). Three d). Four

14. A microwave system requires accurate measurement of a small reflected signal in the presence of a much larger forward signal. Which directional-coupler characteristic should be given the highest priority? [E] / [CO4]

a). Low directivity b). High directivity

c). High insertion loss d). Low isolation

15. Which parameter is common to both rectangular and circular waveguides? [K] / [CO2]

a). Cutoff frequency b). Rectangular dimensions

c). Circular radius d). TE_{10} mode

16. The polarization of an electromagnetic wave is determined by observing the behavior of the: [U] / [CO1]

a). Electric field vector b). Wave velocity only

c). Frequency spectrum only d). Power supply voltage

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. Explain why coaxial transmission lines are widely used in microwave and RF systems, despite their limitations at very high frequencies. [U] / [CO2]

18. Explain the importance of microwave engineering in modern communication systems. [U] / [CO1]

19. Explain any four advantages of microwaves over lower-frequency radio waves. [U] / [CO1]

20. Why is impedance matching required in a transmission line? [U] / [CO2]

21. Explain how electron bunching takes place in the drift space of a multi-cavity klystron. [U] / [CO3]

22. Explain the construction and working principle of an E-plane T-junction. [U] / [CO4]

23. Why are varactor diodes suitable for electronic tuning instead of mechanically variable capacitors in microwave circuits? [U] / [CO3]

24. Explain the use of a Gunn diode as a microwave oscillator. [U] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Explain the advantages and limitations of a TWT compared with a two-cavity klystron. Discuss why TWTs are preferred in applications requiring wide bandwidth. [A] / [CO3]

26. Explain why the phase velocity in a waveguide can be greater than the velocity of light in free space without violating the principles of relativity. [U] / [CO1]

27. List the major advantages and disadvantages of stripline and microstrip lines. [K] / [CO2]

28. What is a waveguide taper? Explain its purpose in microwave transmission systems. [K] / [CO4]

[12x2 = 24]